STAT 5701 **Midterm Exam** **Name:**

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Fall 2024 *parts!* **Uni:**

1. Let the joint pdf for continuous random variables (X, Y) be

$$f(x,y) = cx + 1, \qquad x, y \geq 0, \qquad x + y \leq 1$$

a) Find the constant c.

$$1 = \int_0^1 \int_0^{1-x} (cx+1)dy\,dx = \int_0^1 (cx+1)(1-x)dx$$

$$\int_0^1 (cx + 1 - cx^2 - x)dx = \frac{1}{2} + \frac{c}{6} \Rightarrow c = 3$$

b) Find both marginal pdf's $f_X(x)$ and $f_Y(y)$.
Hint: If you couldn't find the constant c from part a), assume c = 2.

$$f_X(x) = \int_0^{1-x} (3x+1)dy = (3x+1)(1-x), x \in [0,1]$$

$$f_Y(y) = \int_0^{1-y} (3x+1)dx$$

$$= \left(\frac{3x^2}{2} + x\right)\Big]_0^{1-y} = \frac{1}{2}(1-y)(5-3y), y \in [0,1]$$

2. Suppose the pdf of the continuous random variable X is given by:

$$f(x) = \begin{cases} 1 + x, & x \in [-1,0) \\ 1 - x, & x \in [0,1] \\ 0, & x \notin [-1,1] \end{cases}$$

a. Show $f(x)$ is a proper pdf.
Hint: $\int_{-1}^{1} f(x)dx = \int_{-1}^{0} f(x)dx + \int_{0}^{1} f(x)dx$

$$\int_{-1}^{1} f(x)dx = \int_{-1}^{0} (1 + x)dx + \int_{0}^{1} (1 - x)dx$$

$$= x]_{-1}^{0} + \frac{x^2}{2}\Big]_{-1}^{0} + x]_{0}^{1} - \frac{x^2}{2}\Big]_{0}^{1} = 1 - \frac{1}{2} + 1 - \frac{1}{2} = 1$$

b. Derive the cdf $F(x)$.

For $x < -1$, we have $F(x) = 0$.
For $x \in [-1,0)$, we have

$$F(x) = \int_{-1}^{x} (1 + t)dt = t + \frac{t^2}{2}\Big]_{-1}^{x}$$

$$= x - (-1) + \frac{x^2}{2} - \frac{(-1)^2}{2} = x + \frac{x^2}{2} + \frac{1}{2}$$

Finally,

$$F(x) = \begin{cases} x + \frac{x^2}{2} + \frac{1}{2}, x \in [-1,0) \\ \frac{1}{2} + x - \frac{x^2}{2}, x \in [0,1] \end{cases}$$

c. Find P(-1 ≤ X ≤ -0.4).

$$P(-1 \le X \le -0.4) = F(-0.4) - F(-1) = -0.4 + \frac{(-0.4)^2}{2} + \frac{1}{2} = 0.18$$

d. Find E(X).

E(X) = 0 because the distribution is symmetric about 0.
Or, solve with the integral $E(X) = \int_{-1}^{0} x(1 + x)dx + \int_{0}^{1} x(1 - x)dx$

3. The discrete random variables X and Y have a joint probability mass function

$$f(x,y) = \begin{cases} cxy, & \text{for } x = 1,2,3, y = 1,2 \\ 0, & \text{o/w} \end{cases}$$

a) Find the value of the constant c.

$c[1(1) + 1(2) + 2(1) + 2(2) + 3(1) + 3(2)] = 1$
$\Leftrightarrow 18c = 1 \Leftrightarrow c = 1/18$

b) Compute the marginal pmf $f_Y(y)$.

Y	1	2
$p(y)$	$\frac{1(1) + 2(1) + 3(1)}{18} = \frac{1}{3}$	$\frac{1(2) + 2(2) + 3(2)}{18} = \frac{2}{3}$

c) Determine the conditional pmf $f(x \mid y = 1)$.

$X \mid Y = 1$	1	2	3
$p(x \mid y = 1)$	$\frac{\frac{1}{18}}{\frac{1}{3}} = \frac{1}{6}$	$\frac{\frac{2}{18}}{\frac{1}{3}} = \frac{1}{3}$	$\frac{\frac{3}{18}}{\frac{1}{3}} = \frac{1}{2}$

4. A *discrete* random variable X has a *cumulative* distribution function with jumps at the points $x = 1, \ldots, 6$, and given by the formula

$$F(x) = \begin{cases} 0, & x \le 0 \\ \dfrac{x^3}{216}, & x = 1, \ldots, 6 \\ 1, & x > 6 \end{cases}$$

a. Find $P(5 < X \le 6)$

$$P(5 < X \le 6) = F(6) - F(5) = \frac{6^3}{216} - \frac{5^3}{216} = \frac{91}{216} = 0.42$$

b. Find $P(1 < X < 5 \mid 2 \le X < 3)$

$P(1 < X < 5 \mid 2 \le X < 3) = 1$, because $\{2 \le X < 3\} \subset \{1 < X < 5\}$

c. Derive the table with the probability mass function $p(x)$ of X.

x	1	2	3	4	5	6
P(x)	1/216	7/216	19/216	37/216	61/216	91/216

5. Suppose that in the general population, the probability of having a certain common disease (aka base rate) is 0.44. We denote the true presence or absence of the disease as the value of a parameter, θ, that can have the value θ = 1 if disease is present in a person, or the value θ = 0 if the disease is absent. Suppose that there is a test for the disease that has a 87% sensitivity rate, which means that if a person has the disease, then the test result is positive 87% of the time. We denote a positive test result as T = +, and a negative test result as T = −. Then the sensitivity is expressed formally as P(T = + | θ = 1) = 0.87. Suppose also that the test has a false alarm rate of 7%. We denote the false alarm rate as P(T = + | θ = 0) = 0.07.

a) Suppose we sample a person at random from the population, administer the test, and it comes up positive. What is the posterior probability that the person has the disease? Mathematically expressed, we are asking, what is P(θ = 1 |T = +)?

$$P(\theta = 1|T = +)$$
$$= \frac{P(T = +|\theta = 1)P(\theta = 1)}{P(T = +|\theta = 1)P(\theta = 1) + P(T = +|\theta = 0)P(\theta = 0)}$$
$$= \frac{0.87(0.44)}{0.87(0.44) + 0.07(0.56)} = 0.91$$

b) Suppose that the same randomly selected person as in part a) gets re-tested after the first test result was positive, and on the re-test, the result is also positive. Given results of both tests, and assuming conditional independence of the test results, what is the probability that the person has the disease? Mathematically expressed, we are asking, what is P(θ = 1 |T_1 = +, T_2 = +) under the assumption that P(T_1 = +, T_2 = + | θ) = P(T_1 = + | θ)P(T_2 = + | θ).

$$P(\theta = 1|T_1 = +, T_2 = +)$$
$$= \frac{P(T_1 = +, T_2 = +|\theta = 1)P(\theta = 1)}{P(T_1 = +, T_2 = +|\theta = 1)P(\theta = 1) + P(T_1 = +, T_2 = +|\theta = 0)P(\theta = 0)}$$
$$= \frac{(0.87)^2(0.44)}{(0.87)^2(0.44) + (0.07)^2(0.56)} = 0.99$$

c) **Bonus**: Show that under the above assumption of conditional independence we have that:

$$P(\theta = 1|T_1 = +, T_2 = +) = \frac{P(T_2 = +|\theta = 1)P(\theta = 1|T_1 = +)}{P(T_2 = +|\theta = 1)P(\theta = 1|T_1 = +) + P(T_2 = +|\theta = 0)P(\theta = 0|T_1 = +)}$$

$$\frac{P(T_2 = +|\theta = 1)P(\theta = 1|T_1 = +)}{P(T_2 = +|\theta = 1)P(\theta = 1|T_1 = +) + P(T_2 = +|\theta = 0)P(\theta = 0|T_1 = +)}$$
$$= \frac{P(T_2 = +|\theta = 1)P(T_1 = +|\theta = 1)P(\theta = 1)P(T_1 = +)}{P(T_2 = +|\theta = 1)P(T_1 = +|\theta = 1)P(\theta = 1)P(T_1 = +) + P(T_2 = +|\theta = 0)P(T_1 = +|\theta = 0)P(\theta = 0)P(T_1 = +)}$$
$$= \frac{P(T_1 = +, T_2 = +|\theta = 1)P(\theta = 1)}{P(T_1 = +, T_2 = +|\theta = 1)P(\theta = 1) + P(T_1 = +, T_2 = +|\theta = 0)P(\theta = 0)}$$

6. An urn contains eight green marbles and twelve blue marbles. We draw sequentially two marbles from the urn without replacement. We define the events:

   A = {the first marble is blue}
   B = {the second marble is blue}
   C = {both are blue}

   Assume that all marbles are identical except for color and equally likely to be selected at each draw. Find the following probabilities:

   a. P(A).

$$P(A) = \frac{12}{20} = \frac{3}{5} = 0.6$$

   b. P(C).

$$P(C) = \frac{12}{20} \times \frac{11}{19} = \frac{33}{95}$$

   c. P(B).

$$P(B) = \frac{12}{20} \times \frac{11}{19} + \frac{8}{20} \times \frac{12}{19} = \frac{12}{20} = \frac{3}{5} = 0.6$$

   d. P(B $\cup$ $\bar{C}$).

$$P(B \cup \bar{C}) = 1$$

   e. P(A | B).

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(C)}{P(B)} = \frac{\frac{12}{20} \times \frac{11}{19}}{\frac{12}{20}} = \frac{11}{19}$$